

OPERATING SYSTEMS CONCEPT – LAB

EXP – 10: PAGE REPLACEMENT ALGORITHMS – FIFO & LRU

FIFO

```
Activities Terminal May 14 19:51
rathnavelu06@localhost:~/rathna/pageReplacement — /usr/bin/vim fifo.c

#include <stdio.h>
int main(){
    int i, j, n, a[50], frame[10], no, k, avail, count=0;
    printf("\nEnter number of pages: ");
    scanf("%d", &n);
    printf("Enter the page numbers (reference string): \n");
    for(i=1; i<=n; i++){
        scanf("%d", &a[i]);
    }
    printf("Enter number of frames: ");
    scanf("%d", &no);
    for(i=0; i<no; i++){
        frame[i]=-1;
    }
    j=0;
    printf("\nRef String\tPage frames\n");
    printf("-----\t-----\n");
    for(i=1; i<=n; i++){
        printf("%d\t\t", a[i]);
        avail=0;
        for(k=0; k<no; k++){
            if(frame[k]==a[i]){
                avail=1;
            }
        }
        if(avail==0){
            frame[j]=a[i];
            j=(j+1)%no;
            count++;
            for(k=0; k<no; k++){
                if(frame[k]==-1){
                    printf("%d ", frame[k]);
                }
                else{
                    printf("- ");
                }
            }
        }
        else{
            printf(" (Hit)");
        }
        printf("\n");
    }
    printf("\nTotal page faults: %d\n", count);
    return 0;
}

"fifo.c" 47L, 855B written 47,0-1 All
```

```
Activities Terminal May 14 19:53
rathnavelu06@localhost:~/rathna/pageReplacement

[rathnavelu06@localhost pageReplacement]$ vi fifo.c
[rathnavelu06@localhost pageReplacement]$ gcc fifo.c -o fifo
[rathnavelu06@localhost pageReplacement]$ ./fifo

Enter number of pages: 7
Enter the page numbers (reference string):
1 3 0 3 5 6 3
Enter number of frames: 3

Ref String      Page frames
-----
1              1 - -
3              1 3 -
0              1 3 0
3              (Hit)
5              5 3 0
6              5 6 0
3              5 6 3

Total page faults: 6
[rathnavelu06@localhost pageReplacement]$
```

VI, PWD, TOUCH, CAT COMMANDS

```
Activities Terminal May 14 20:13
rathnavelu06@localhost:~/rathna/pageReplacement — /usr/bin/vim lru.c

#include <stdio.h>
int findLRU(int time[], int n){
    int i, minimum=time[0], pos=0;
    for(i=1; i<n; i++){
        if(time[i]<minimum){
            minimum=time[i];
            pos=i;
        }
    }
    return pos;
}

int main(){
    int no_of_frames, no_of_pages, frames[10], pages[30], counter=0, time[10];
    int flag1, flag2, i, j, pos, faults=0;
    printf("Enter number of frames: ");
    scanf("%d", &no_of_frames);
    printf("Enter number of pages: ");
    scanf("%d", &no_of_pages);
    printf("Enter reference string: ");
    for(i=0; i<no_of_pages; i++){
        scanf("%d", &pages[i]);
    }
    for(i=0; i<no_of_frames; i++){
        frames[i]=-1;
    }
    printf("\nRef String | Page Frames\n");
    printf("-----|-----\n");
    for(i=0; i<no_of_pages; i++){
        flag1=flag2=0;
        for(j=0; j<no_of_frames; j++){
            if(frames[j]==pages[i]){
                counter++;
                time[j]=counter;
                flag1=flag2=1;
                break;
            }
        }
        if(flag1==0){
            for(j=0; j<no_of_frames; j++){
                if(frames[j]==-1){
                    counter++;
                    faults++;
                    frames[j]=pages[i];
                    time[j]=counter;
                    flag2=1;
                    break;
                }
            }
        }
        if(flag2==0){
            pos=findLRU(time, no_of_frames);
            counter++;
            faults++;
            frames[pos]=pages[i];
            time[pos]=counter;
        }
        printf("      %d      |", pages[i]);
        for(j=0; j<no_of_frames; j++){
            if(frames[j]!=-1){
                printf("%d ", frames[j]);
            }
            else{
                printf("- ");
            }
        }
        printf("\n");
    }
    printf("\nTotal Page Faults = %d\n", faults);
    return 0;
}
```

```
Activities Terminal May 14 20:15
rathnavelu06@localhost:~/rathna/pageReplacement

[rathnavelu06@localhost pageReplacement]$ vi lru.c
[rathnavelu06@localhost pageReplacement]$ gcc lru.c -o lru
[rathnavelu06@localhost pageReplacement]$ ./lru
Enter number of frames: 3
Enter number of pages: 7
Enter reference string: 7 0 1 2 0 3 0

Ref String | Page Frames
-----|-----
      7      | 7 - -
      0      | 7 0 -
      1      | 7 0 1
      2      | 2 0 1
      0      | 2 0 1
      3      | 2 0 3
      0      | 2 0 3

Total Page Faults = 5
[rathnavelu06@localhost pageReplacement]$
```